

# MH2500 Probability and Introduction to Statistics

## Slides for Week 6-1

Topics: More on normal variables and exponential variables

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## Theorem:

Let  $X$  be a continuous random variable with probability density function  $f(x)$ , then for any real-valued function  $g$ ,

$$\mathbb{E}(g(X)) = \int_{-\infty}^{\infty} g(x)f(x)dx,$$

provided that  $\int_{-\infty}^{\infty} |g(x)|f(x)dx$  exists.

We will prove it on Friday, when we will have more time.

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For  $Y = aX + b$ ,

$$\begin{aligned}\mathbb{E}(aX + b) &= \int_{-\infty}^{\infty} (ax + b)f(x)dx \\ &= a \int_{-\infty}^{\infty} xf(x)dx + b \int_{-\infty}^{\infty} f(x)dx = a\mathbb{E}(x) + b.\end{aligned}$$

## Example

The density function of  $X$  is given by

$$f(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1; \\ 0 & \text{otherwise.} \end{cases}$$

Find  $\mathbb{E}(e^X)$ .

By

$$\mathbb{E}(e^X) = \int_0^1 e^x dx = e^x \Big|_0^1 = e - 1.$$

.....  
We can also obtain it by applying the definition.

Let  $Y = e^X$ . Then the cumulative distribution function  $F_Y$  is: for  $1 \leq x \leq e$  (so  $0 \leq \log x \leq 1$ ),

$$\begin{aligned} F_Y(x) &= P\{Y \leq x\} = P\{e^X \leq x\} = P\{X \leq \log x\} \\ &= \int_0^{\log x} f(t) dt = \int_0^{\log x} 1 dt = \log x. \end{aligned}$$

Thus,

$$f_Y(x) = (F_Y(x))' = \frac{1}{x}$$

for  $1 \leq x \leq e$ . This gives

$$\mathbb{E}(e^X) = \mathbb{E}(Y) = \int_{-\infty}^{\infty} x f_Y(x) dx = \int_1^e x \left( \frac{1}{x} \right) dx = \int_1^e 1 dx = e - 1.$$

Same result, of course.

# Normal Approximation to Binomial Distribution

## DeMoivre-Laplace Limit Theorem

Let  $p$  be a constant and  $S_n \sim \text{Bin}(n, p)$ , then for any  $a < b$ ,

$$\lim_{n \rightarrow \infty} P \left\{ a \leq \frac{S_n - np}{\sqrt{np(1-p)}} \leq b \right\} = \Phi(b) - \Phi(a).$$

Two ways of approximating a binomial distribution

- Poisson:  $n$  large ( $n \geq 50$ ),  $p$  small ( $np < 5$ ).
- Normal:  $np(1-p)$  large (good enough when  $np(1-p) \geq 10$ )

## Example

Flip a fair coin 40 times and let  $X$  be the number of heads. Find the probability that  $X = 20$ .

Use  $P\{X = i\} = P\{i - 1/2 < X < i + 1/2\}$  — change to continuous.

$$\begin{aligned}P\{X = 20\} &= P\{19.5 < X < 20.5\} \\&= P\left\{\frac{19.5 - 20}{\sqrt{10}} < \frac{X - 20}{\sqrt{10}} < \frac{20.5 - 20}{\sqrt{10}}\right\} \\&\approx P\left\{-0.16 < \frac{X - 20}{\sqrt{10}} < 0.16\right\} \\&= \Phi(0.16) - \Phi(-0.16) = 2\Phi(0.16) - 1 \approx 0.1272.\end{aligned}$$

The exact result, by Binomial distribution, is

$$\binom{40}{20} \left(\frac{1}{2}\right)^{20} \left(\frac{1}{2}\right)^{20} = \binom{40}{20} \left(\frac{1}{2}\right)^{40} \approx 0.1254.$$

## Example

The ideal size of a class at a college is 150 students. The college knows only 30% of those who were given an admission offer will actually attend. The college makes 450 offers of admission this year. Compute the probability that more than 150 students will attend this college.

**Solution:** Let  $X$  be the number of students who will attend, then  $X \sim \text{Bin}(450, 0.3)$  (assuming independence!). Thus

$$\begin{aligned} P\{X \geq 151\} &= P\{X \geq 150.5\} \quad (\text{including } 151) \\ &= P\left\{ \frac{X - (450)(0.3)}{\sqrt{450(0.3)(0.7)}} \geq \frac{150.5 - (450)(0.3)}{\sqrt{450(0.3)(0.7)}} \right\} \\ &= P(Z \geq 1.59) \approx 1 - \Phi(1.59) \approx 0.0559. \end{aligned}$$

## Example

In 10,000 independent tosses of a coin, the coin landed on heads 5800 times. Is it reasonable to assume that the coin is not fair? Explain.

**Solution:** Suppose that the coin is fair and  $X$  is the number of heads, then  $X \sim \text{Bin}(10000, 0.5)$  and so  $\mathbb{E}(X) = 5000$  and  $\text{Var}(X) = 2500$ . Let us calculate the probability  $P\{X \geq 5800\}$ .

$$\begin{aligned} P\{X \geq 5800\} &= P\{X \geq 5799.5\} \approx P_{Z \sim N(0,1)} \left\{ Z \geq \frac{5799.5 - 5000}{\sqrt{2500}} \right\} \\ &= P_{Z \sim N(0,1)} \left\{ Z \geq \frac{799.5}{50} \right\} = P_{Z \sim N(0,1)} \{Z \geq 15.99\}, \end{aligned}$$

which is a tiny probability (almost 0). That means, it is extremely **unlikely** (nearly impossible) to see 5800 heads with a fair coin in practice; and thus it is reasonable to assume that the coin is not fair.



# Exponential Distribution

The exponential density function is

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & \text{if } x \geq 0; \\ 0, & \text{if } x < 0. \end{cases}$$

The cumulative distribution function is

$$F(x) = \int_{-\infty}^x f(t) dt = \int_{-\infty}^0 0 dt + \int_0^x \lambda e^{-\lambda t} dt = \begin{cases} 1 - e^{-\lambda x}, & \text{if } x \geq 0; \\ 0, & \text{if } x < 0. \end{cases}$$

$$P(X > x) = 1 - F(x) = 1 - (1 - e^{-\lambda x}) = e^{-\lambda x}, \quad x \geq 0.$$

Like the Poisson distribution, the exponential density depends on a single parameter  $\lambda > 0$ .

# Expectation and Variance the Exponential distribution

Let  $X \sim \text{Exp}(\lambda)$ . Find  $\mathbb{E}(X)$  and  $\text{Var}(X)$ .

$$\begin{aligned}\mathbb{E}(X^n) &= \int_0^\infty x^n \lambda e^{-\lambda x} dx \\&= -e^{-\lambda x} x^n \Big|_{x=0}^\infty + \int_0^\infty e^{-\lambda x} n x^{n-1} dx \\&= \frac{n}{\lambda} \int_0^\infty x^{n-1} \lambda e^{-\lambda x} dx \\&= \frac{n}{\lambda} \mathbb{E}(X^{n-1}).\end{aligned}$$

$$\mathbb{E}(X) = \frac{1}{\lambda}, \quad \text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E} X)^2 = \frac{2}{\lambda} \cdot \frac{1}{\lambda} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}$$

# Properties of the Exponential distribution

- Related to Poisson.

- It is used to model **between the occurrence of events** in an interval of time.
- E.g. The time between phone calls received at a help center,  
The time between failures of a system, etc,  
In general, “the time between two consecutive events **where the occurrence of events follow a Poisson distribution**”.

- Memoryless.

- The memoryless character of the exponential distribution follows from its relation to a Poisson process.

# Exponential: Relation with Poisson

Suppose events occur in unit time as a Poisson process with parameter  $\lambda$  and that an event occurs at time  $t_0$ .

(Recall: If  $X$  is the number of events in unit time,  $P\{X = k\} = \frac{\lambda^k}{k!} e^{-\lambda}$ , and so  $P\{X = 0\} = e^{-\lambda}$ .)

Instead studying the probability of the number of events in a certain time, we consider  $T$  **the length of time until the next event**. Then

$$P\{T > t\} = P\{\text{no events in } (t_0, t_0 + t)\} = P\{Y = 0\},$$

where  $Y$  is the number of events during time  $(t_0, t_0 + t)$ , (i.e., in time  $t$ ).

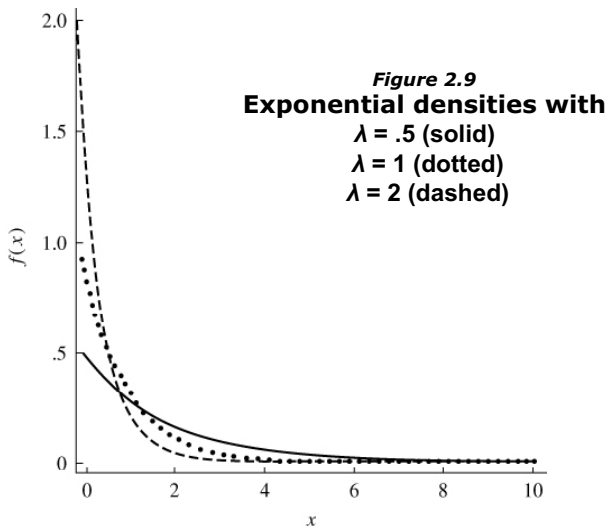
Then  $Y$  is a Poisson process with parameter  $\lambda t$ . Hence

$$P\{T > t\} = P\{Y = 0\} = e^{-\lambda t}.$$

i.e., 
$$F(t) = P(0 \leq T \leq t) = 1 - e^{-\lambda t}.$$

# Comparing different values of $\lambda$

Note that as  $\lambda$  increases, the density decays (drops off) more rapidly.



## Example

Suppose  $T$  is the lifetime of an electronic component. We model  $T$  using an exponential distribution with parameter  $\lambda$ .

Suppose the electronic component has lasted a length of time  $s$ .

Find the probability that it will last at least another length of time  $t$ .

**Solution:**

$$\begin{aligned} P\{T \geq t + s | T \geq s\} &= \frac{P\{T > t + s \text{ and } T > s\}}{P\{T > s\}} \\ &= \frac{P\{T > t + s\}}{P\{T > s\}} = \frac{e^{-\lambda(t+s)}}{e^{-\lambda s}} = e^{-\lambda t}. \end{aligned}$$

# Exponential: Memoryless

Exponential distribution is **memoryless**:

$$P\{T \geq t + s | T \geq s\} = P\{T \geq t\}.$$

The exponential distribution is **not a good model for human lifetime**.

E.g. the probability that a 20 year old will live at least another 10 years is not the same as the probability that an 80 year old will live at least another 10 years.

It can be shown that a memoryless distribution follows an **exponential distribution** in the continuous case and **geometric distribution** in the discrete case.